BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics (Objective) Group - I

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 7475

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. If the current in a wire which is placed perpendicular to a magnetic field increases, the force on the wire:

(A) Increases
(B) Decreases
(C) Remains the constant
(D) Will be zero

اگر میگنیٹک فیلڈ میں عموداً رکھی ہوئی تار میں سے بہنے والے کرنٹ کی مقدار بڑھ جائے تو اس پر عمل کرنے والی میگنیٹک فورس:

(A) بڑھے گی
(B) کم ہوگی
(C) [illegible in source scan]
(D) صفر ہوگی

Q2. An energy-saver light bulb uses energy:

(A) 11 J
(B) 12 J
(C) 13 J
(D) 15 J

انرجی سیور بلب توانائی استعمال کرتا ہے

Q3. Capacitance is defined as:

(A) VC
(B) Q/V
(C) QV
(D) V/Q

کپیسٹنس کی تعریف اس طرح کی جاتی ہے

Q4. Which type of image is formed by a convex lens on a screen?

(A) Inverted and real
(B) Inverted and virtual
(C) Upright and real
(D) Upright and virtual

کونویکس لینس سکرین پر کس قسم کی امیج بناتا ہے؟

(A) الٹی اور حقیقی
(B) الٹی اور مجازی
(C) سیدھی اور حقیقی
(D) سیدھی اور مجازی

Q5. Which is an example of longitudinal waves?

(A) Sound waves
(B) Light waves
(C) Radio waves
(D) Water waves

کون سی لونگیٹیوڈنل ویوز کی مثال ہے؟

(A) آواز کی لہریں
(B) روشنی کی لہریں
(C) ریڈیو لہریں
(D) پانی کی لہریں

Q6. Which characteristic of a wave is independent of the others?

(A) Speed
(B) Frequency
(C) Amplitude
(D) Wavelength

لہر کی کون سی خصوصیت دوسروں سے آزاد ہوتی ہے؟

(A) رفتار
(B) فریکوئنسی
(C) ایمپلیٹیوڈ
(D) ویولینتھ

Q7. Half life of $^{131}_{53}I$ is:

(A) 12.3 years
(B) 8.07 days
(C) 12.3 days
(D) 8.07 year

آئیوڈین $^{131}_{53}$ کی ہاف لائف ہے:

(A) 12.3 سال
(B) 8.07 دن
(C) 12.3 دن
(D) 8.07 سال

Q8. How many basic operations of digital electronics are?

(A) 3
(B) 4
(C) 5
(D) 2

ڈیجیٹل الیکٹرانکس کے بنیادی آپریشنز کتنے ہیں؟

Q9. If the object is on the right side of lens, then p is:

(A) Positive
(B) Negative
(C) Smaller
(D) Larger

اگر آبجیکٹ لینس کے دائیں جانب ہو تو p کی قیمت ہوگی:

(A) مثبت
(B) منفی
(C) چھوٹی
(D) بڑی

Q10. Isotopes are atoms of same element with different:

(A) Atomic mass
(B) Atomic number
(C) Number of proton
(D) Number of electron

آئسوٹوپس ایک ہی عنصر کے ایسے ایٹم ہوتے ہیں جن کا مختلف ہوتا ہے:

(A) ایٹمی ماس
(B) ایٹمی نمبر
(C) پروٹان کی تعداد
(D) الیکٹران کی تعداد

Q11. What does the term e-mail stand for?

(A) Emergency mail
(B) Electronic mail
(C) Extra mail
(D) External mail

ای میل سے کیا مراد ہے؟

(A) ایمرجنسی میل
(B) الیکٹرانک میل
(C) ایکسٹرا میل
(D) ایکسٹرنل میل

Q12. The output of a two input NOR gate is 1 when:

(A) A = 1 and B = 0
(B) A = 0 and B = 1
(C) A = 0 and B = 0
(D) A = 1 and B = 1

دو ان پٹ والے NOR گیٹ کا آؤٹ پٹ 1 ہوتا ہے جب:

Physics (Subjective) Group - I

Time Allowed: 01:45 Hours	**Maximum Marks:** 48	**Paper Code:** 7475

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) Define restoring force and write its formula.

(ii) Define the pitch of sound.

(iii) Prove that $v = f \lambda$

(iv) For spherical mirror, define radius of curvature and focal length

(v) Define total internal reflection.

(vi) Write the audible sound frequency range.

(vii) Define electromagnetic waves.

(viii) What is lens formula? Write it.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Define electric current. Write its SI unit also.

(ii) How the potential difference across a circuit component is measured? Draw its diagram.

(iii) Define the term kilowatt-hour. How much Joules are in one kilowatt-hour?

(iv) State the rule to find the direction of magnetic field around the current carrying conductor

(v) In mutual induction let the current in first coil becomes steady. Write about induced current in second coil.

(vi) Write the names of four logic gates.

(vii) Which logic gate is also called inverter? Draw its symbol also.

(viii) Draw the circuit diagram of AND-gate.

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) How a capacitor stores a charge?

(ii) Define tuning circuit.

(iii) State Coulomb's law and write formula for finding force

(iv) What is the difference between data and information?

(v) What is meant by software?

(vi) How light signals are sent through optical fiber?

(vii) What is meant by unstable nuclei?

(viii) Define nuclear fusion.

> Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

> حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کریں۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 04) Define simple pendulum. Prove that motion of a simple pendulum is simple harmonic motion also write down its time period equation.

Q5(b). (Marks 05) A capacitor holds 0.06 Coulombs of charge when fully charged by a 9 volt battery. Calculate capacitance of the capacitor.

Q6(a). (Marks 04) Define electromagnetic induction and explain with example.

Q6(b). (Marks 05) An object is 30cm tall located 10.5cm from a concave mirror with focal length 16cm: (i) Where is image located? (ii) How high is it?

Q7(a). (Marks 04) Define radioisotopes and write its two uses.

Q7(b). (Marks 05) By applying a potential difference of 10V across a conductor, a current of 1.5A passes through it. How much energy would be obtained from the current in 2 minutes?

GROUP – II

Physics (Objective) Group - II

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 7478

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The number of neutrons in Tritium $^{3}_{1}H$:

(A) 1 (B) 2

(C) 3 (D) 4

ٹرائیٹیم $^{3}_{1}H$ میں نیوٹران کی تعداد ہے:

Q2. During natural radioactivity, how many types of radiations are emitted?

(A) 1 (B) 2

(C) 3 (D) 4

قدرتی ریڈیو ایکٹیویٹی کے دوران کتنی اقسام کی شعاعیں خارج ہوتی ہیں؟

Q3. The half life of a certain isotope is 1 day. What is the quantity of the isotope after 2 days?

(A) 1/2 (B) 1/4

(C) 1/8 (D) None of these

[illegible in source scan]

Q4. The number of important components of Computer Based Information System (CBIS) is:

(A) 2 (B) 3

(C) 5 (D) 6

کمپیوٹر بیسڈ انفارمیشن سسٹم کے اہم اجزاء کی تعداد ہے:

Q5. NOT gate is also called:

(A) Conductor (B) Amplifier

(C) Transistor (D) Invertor

NOT گیٹ کو یہ بھی کہتے ہیں:

Q6. SI unit of electric power is:

(A) Joule
(B) Watt
(C) Volt
(D) Ampere

الیکٹرک پاور کا ایس آئی یونٹ ہے:

Q7. One micro Coulomb is equal to:

(A) $10^{-3}C$
(B) $10^{3}C$
(C) $10^{-6}C$
(D) $10^{6}C$

ایک مائیکرو کولمب برابر ہے:

Q8. SI unit of Coulomb's constant k is:

(A) $Nm^{-2}C^{2}$
(B) $Nm^{-2}C^{-2}$
(C) $Nm^{2}C^{2}$
(D) $Nm^{2}C^{-2}$

کولمب کے مستقل k کا ایس آئی یونٹ ہے:

Q9. The types of spherical mirror are:

(A) 2
(B) 3
(C) 4
(D) 5

سفیریکل مرر کی اقسام ہیں:

Q10. Refractive index is equal to:

(A) n = C/V
(B) n = CV
(C) n = V/C
(D) n = I/CV

ریفریکٹیو انڈیکس برابر ہے:

Q11. The time period of vibrating mass spring, when its mass is doubled:

(A) Will remain the same
(B) Will remain half
(C) Will be increased
(D) Will be decreased

وائبریٹنگ ماس اسپرنگ کا ٹائم پیریڈ، جب اس کا ماس دگنا کر دیا جائے:

(A) ایک جیسی رہے گی
(B) آدھی رہ جائے گی
(C) بڑھ جائے گی
(D) کم ہو جائے گی

Q12. The unit of time period is:

(A) Hertz
(B) Second
(C) Second / Joule
(D) Second / Coulomb

ٹائم پیریڈ کا یونٹ ہے:

(A) ہرٹز
(B) سیکنڈ
(C) سیکنڈ / جول
(D) سیکنڈ / کولمب

Physics (Subjective) Group - II

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 7478

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers in any FIVE parts.

(i) Define wave.

(ii) A wave moves on a slinky with frequency of 4Hz and wavelength of 0.4m. Find speed of wave.

(iii) Why longitudinal waves move faster than transverse waves in solids?

(iv) Write equation to find speed of sound and also write one factor which affects speed of sound.

(v) What is sound? How it is produced?

(vi) Differentiate between concave and convex mirror.

(vii) Define angle of incidence and angle of reflection.

(viii) Write sign conventions for lenses for focal length.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Why ammeter is connected in series to measure current in circuit?

(ii) Write names of sources of emf.

(iii) Write two features of parallel combination of resistors.

(iv) How NOR gate is made?

(v) Draw the symbol diagram of OR gate and write its Boolean expression.

(vi) Make the truth table for NAND gate.

(vii) Write two uses of MRI in medical field.

(viii) What was the work of Hans Christian Oersted and Ampere in electromagnetic induction?

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) Define Coulomb's law and write its equation.

(ii) Write the formula for the equivalent capacitance for a series combination of a number of capacitors.

(iii) Define Farad.

(iv) Write two advantages of electronic mail.

(v) What is the difference between data and information?

(vi) How are light signals sent through optical fibers?

(vii) What is meant by background radiations?

(viii) What is meant by half life?

> Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

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Q5(a). (Marks 04) Define loudness. On what factors does the loudness depends? Explain.

Q5(b). (Marks 05) A point charge of +2C is transferred from a point at potential 100V to a point at potential 50V. What would be the energy supplied by the charge?

Q6(a). (Marks 04) What is transformer? Explain its working.

Q6(b). (Marks 05) An object 4cm high is placed at a distance of 12cm from a convex lens of focal length 8cm. Calculate the position and size of image.

Q7(a). (Marks 04) What is meant by nuclear fission reaction? With the help of equation explain about the energy released in each fission event.

Q7(b). (Marks 05) Two resistances of 6kΩ and 12kΩ are connected in parallel. A 6V battery is connected across its ends. Find the values of following quantities: (i) Equivalent resistance of parallel combination (ii) Current passing through each of the resistance.